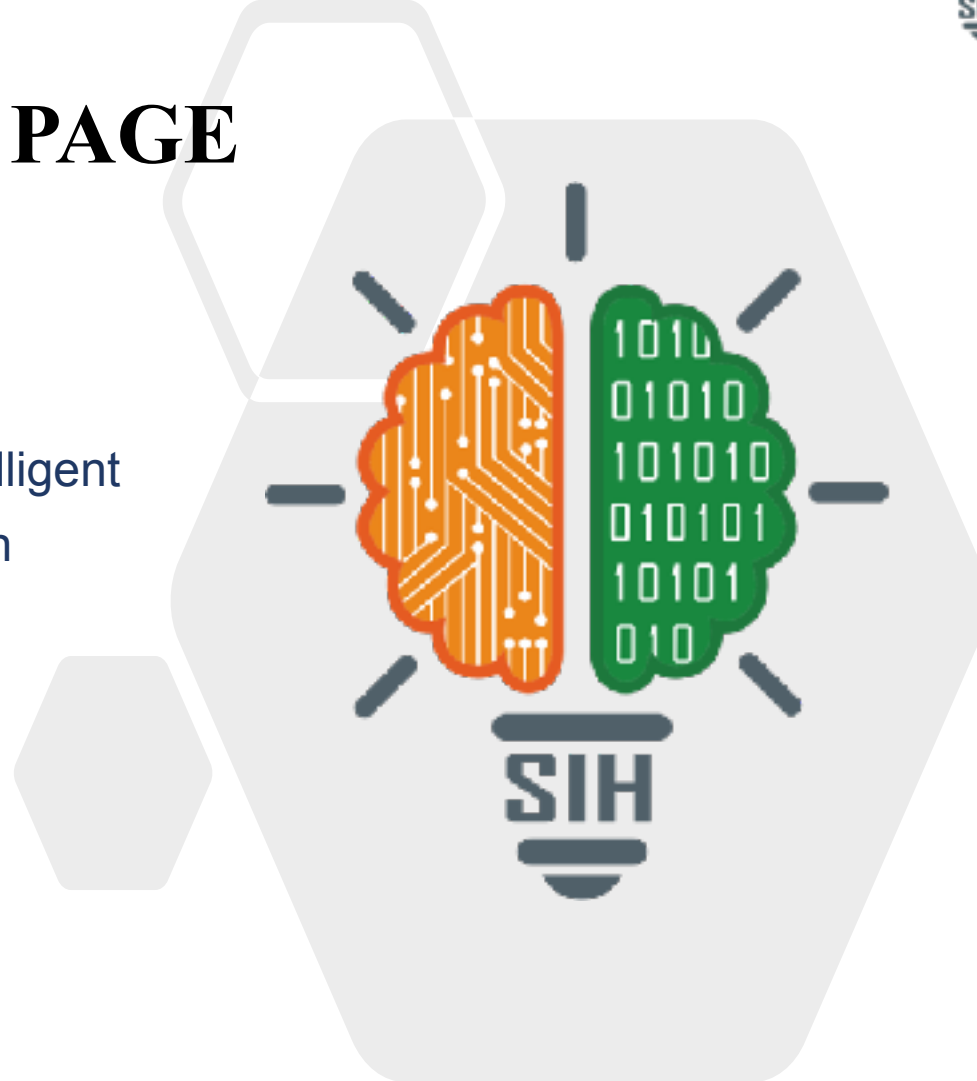


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26168
- **Problem Statement Title** – AI-ML based Intelligent Dead Reckoning system for seamless navigation
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – <Team ID from portal>
- **Team Name (Registered on portal)** – KANABI



IDEA: Intelligent Dead Reckoning (IDR) + GNSS Fusion

Proposed solution

- An edge-deployable engine + Android app that keeps lane-level position when GNSS is lost (tunnels, underpasses, parking, urban canyons, jamming) using only the phone's accelerometer, gyroscope, magnetometer and an offline OpenStreetMap graph. No OBD-II, no wheel sensor, no fixed mount.
- Hybrid AI + physics pipeline: in-vehicle alignment & calibration → AI speed/vibration filter (1-D CNN) → bias-corrected INS → map-matching with non-holonomic constraints → AI-augmented error-state Kalman fusion with instant GNSS-deficit switching.
- Training happens on desktop (IO-VNBD); the engine runs on the phone at sensor rate (pure Dart, no server) and the same core targets the edge engine at ~200 Hz for external/FOG IMUs; the 32 k-param CNN ships via ONNX/TFLite.

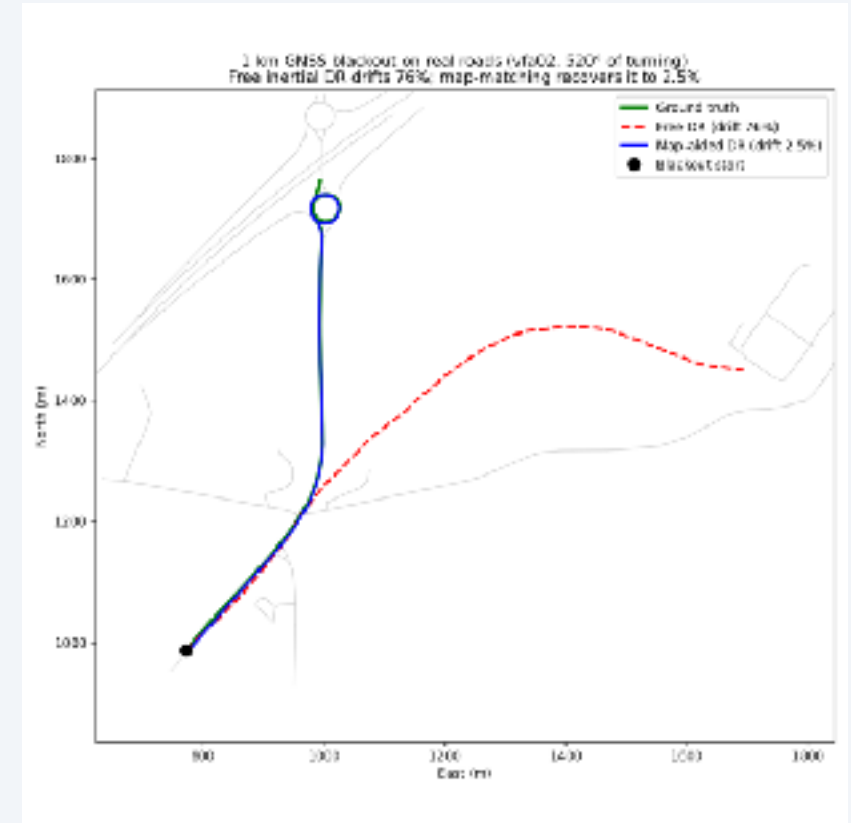
How it addresses the problem

- GNSS drop → seamless switch to inertial tracking within ms; GNSS return → re-converge to GNSS+INS.
- Speed without OBD: last-GNSS speed anchor + learned IMU corrections; vibration, potholes and idling filtered out.
- Heading drift (the real killer of phone DR) is cancelled by snapping to the road graph on every edge.

Innovation and uniqueness

- Data-driven design: on IO-VNBD we measured that speed holds to 1–2 % but gyro heading drifts 19–28° in 30–60 s, so we spend the AI budget where the error really is.
- Mount-agnostic calibration (gravity + PCA of driving acceleration) and one shared core engine for phone and edge/FOG sensors.

Preliminary result on IO-VNBD (drive vfa02): 1 km GNSS blackout, 520° of turning



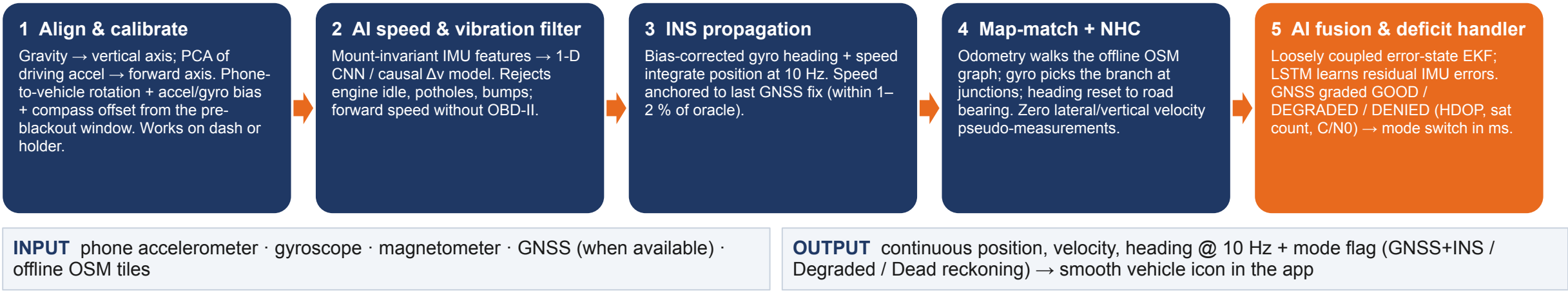
Plain phone IMU drifts 76 % of distance; **our map-aided IDR finishes 2.5 % off** and tracks the roundabout. Benchmark: < 10 %.



Technologies



Methodology: on-device pipeline (10 Hz on phone, ~200 Hz on edge engine)



Workflow and current status

Phase 1 (desktop, done): IO-VNBD loader & synchronisation audit, feature engineering, speed-filter & LSTM training, simulated-blackout evaluation over 369 one-km blackouts, OSM map-matching, browser replay demo.

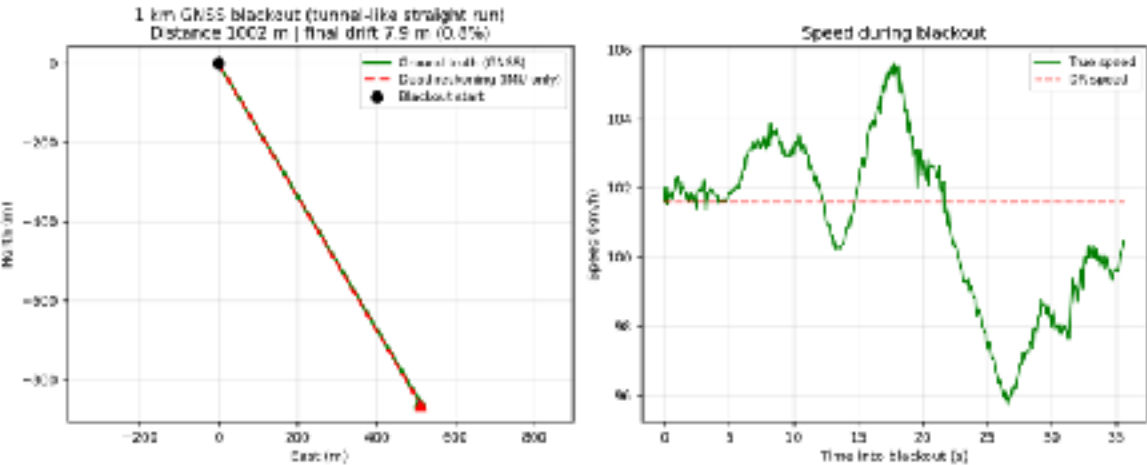
Phase 2 (on-device, working): Android app runs alignment, pre-blackout calibration, dead reckoning and OSM map-matching on the phone (unit-tested on synthetic drives; APK in repo), with a simulate-tunnel switch and live drift read-out on GNSS return. Next: on-device ES-EKF fusion, 200 Hz edge build, real tunnel test.





Feasibility: measured on IO-VNBD, smartphone IMU only

Scenario (simulated GNSS blackout on real drives)	PS benchmark	Achieved
50 m blackout, motorway (< 1 min)	< 5 m	0.4 m (0.7 %)
1 km straight, tunnel-like run @ ~100 km/h	< 100 m	7.9 m (0.8 %)
1 km real roads with turns, map-aided (median of 369)	< 10 %	3.3 % hwy · 4.8–7.0 % urban
Share of 1 km blackouts under 10 % (free DR → map-aided)	—	17 % → 74 %



Position plot inferred from an IO-VNBD test drive: 1 km blackout, final drift 7.9 m. Trained within-vehicle (Driver E), tested on held-out drives vfa02 / vt5 / vw2. Reproducible: python -m src.idr.evaluate / evaluate_mapmatch.

Compute: 32 k-parameter CNN + EKF runs comfortably at 10 Hz on a mid-range Android phone; OSM graph for a city fits in a few MB offline.

Challenges, risks and how we handle them

Challenge / risk	Strategy
Gyro heading drifts 19–28° in 30–60 s → free DR > 10 % beyond ~200 m	Map-matching + NHC reset heading on every road edge; ZUPT at stops; magnetometer only as a weak prior
Absolute speed from 10 Hz vibration aliases (> 5 Hz Nyquist)	Anchor to last GNSS speed + learned Δv correction; 100–200 Hz sampling on phone/edge for the filter
Wrong branch at ambiguous junctions (remaining urban misses)	Multi-hypothesis HMM map-matching with gyro-turn likelihood; prune on GNSS return
Phone mount changes / hand-held, two-wheelers	Continuous re-alignment on detected mount change; vehicle-class model heads
Different IMUs (phone vs FOG) and finale datasets	Sensor-agnostic core; calibration stage absorbs bias/scale; same code path for edge engine
On-device latency and battery	Tiny CNN, TFLite int8, 10 Hz loop; heavy training stays on desktop

Viability

- Zero hardware cost: runs on phones already on every dashboard; no OBD dongle, no vehicle integration.
- Open data (IO-VNBD) + open maps (OSM) → no licensing barrier; models retrain as finale datasets arrive.
- Working repo, reproducible pipeline, replay demo and Android APK already exist.

Who it helps

Delivery & ride-hailing riders

Two-wheelers and cabs with only a phone: no frozen map in flyovers, underpasses and metro tunnels.

Trucks, fleets & older cars

Vehicles without factory INS or OBD feed get continuous tracking and correct turn-by-turn guidance.

Emergency responders

Ambulances and fire units keep position in parking decks and canyons where seconds matter.

Navigation & telematics apps

Drop-in engine for MapmyIndia / fleet SDKs; edge engine for ISRO, defence and rail use with FOG IMUs.

Benchmark the judges set vs. what we already measure

< 10 %

PS target drift

0.7 %

50 m blackout

0.8 %

1 km tunnel run

3.3–7 %

1 km real roads
(median)

Benefits

Social / safety

No missed exits or erratic jumps in tunnels; safer lane guidance for two-wheelers; reliable emergency response in GNSS-denied places.

Economic

Zero added hardware; fewer failed deliveries and detours; one engine for millions of existing phones and fleet devices.

Environmental

Fewer wrong turns and re-routing loops → less fuel and emissions per trip.

Strategic

Resilience against jamming/spoofing and signal loss; complements NavIC; same AI core deploys on edge hardware for space, defence and rail.



Live replay demo + Android app

Real IO-VNBD drive through a 1 km blackout: plain IMU 942 m off, IDR + map-matching 7 m. App shows a smooth vehicle icon with an unmistakable GNSS / Dead-reckoning banner.

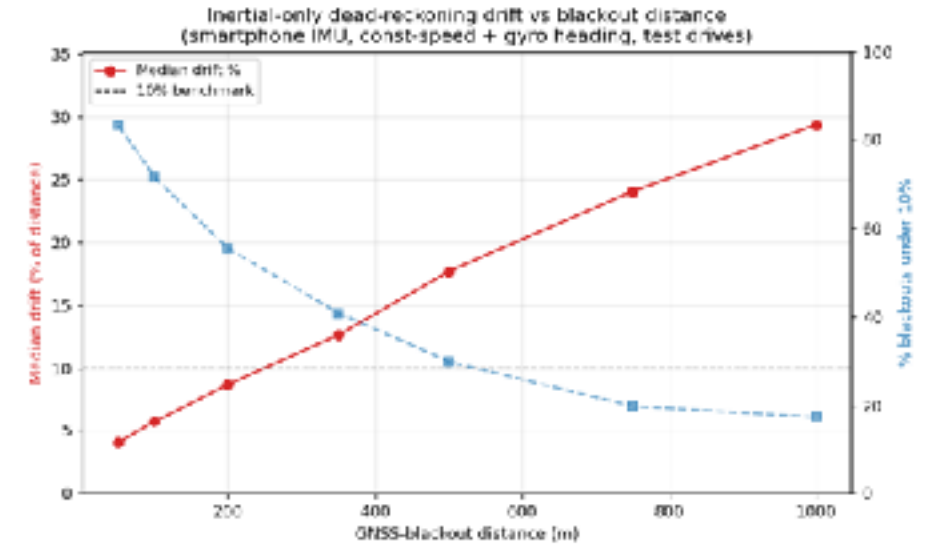
Our work (open for review)

- **Code, models, results:** <https://github.com/AnushtupGhosh5/sih>
- Reproducible pipeline (requirements.txt / Docker): IO-VNBD loader → speed filter training → blackout evaluation → map-matching → figures; RESULTS.md holds every number on this deck.
- Browser replay demo (demo/index.html) and Android APK (mobile_app/ui) included in the repository.

References

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- Brossard M., Barrau A., Bonnabel S. "AI-IMU Dead-Reckoning." IEEE Trans. Intelligent Vehicles, 2020.
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- OpenStreetMap contributors, openstreetmap.org; Boeing G. "OSMnx" (Computers, Environment and Urban Systems, 2017).
- TensorFlow Lite / ONNX Runtime documentation for on-device inference.

Key findings from our IO-VNBD study



- Inertial-only phone DR stays under 10 % only to ~150–200 m; map constraints extend this to 1 km (median 29.3 % → 3.3–7.0 %).
- Holding the last GNSS speed is within 1–2 % of oracle speed over a tunnel; gyro heading, not speed, dominates error.
- Inertial alignment of the dataset (phone vs ECU, 20 sessions) raised Δv correlation from 0.08 to 0.31, proving recoverable motion information in the phone IMU.
- Dataset audit found and fixed a gyro/gravity axis-matching bug that corrupts heading in naive loaders.